

# Spark2 SDK: Repository Structure and API

- Language: C++20

## Repository Structure

```
spark2_sdk/  
|-- CMakeLists.txt  
|-- include/  
|   |-- spark2.h          (main SDK class API)  
|   |-- configurator.h    (configuration API)  
|   |-- kinematics.h      (kinematics API)  
|   |-- types.h           (shared data types/enums)  
|-- examples/  
|   |-- move_point.cpp  
|   |-- move_path.cpp  
|   |-- teach_and_playback.cpp  
|   |-- set_configuration.cpp  
|   |-- calculate_kinematics.cpp
```

## Core SDK API (Spark2)

| API                                              | Description                                                      | Group         |
|--------------------------------------------------|------------------------------------------------------------------|---------------|
| <code>Spark2(config_prefix_path="")</code>       | Create SDK client; optional path prefix for configuration files. | constructor   |
| <code>start(), stop()</code>                     | Start or stop arm runtime session.                               | connection    |
| <code>enableArmJoint(arm_state)</code>           | Enable arm joints by 6-element boolean state array.              | connection    |
| <code>setArmSmoothingMethod(method)</code>       | Select smoothing strategy for single-target arm motion.          | motion mode   |
| <code>movePos(arm_pos, v=50, t=0)</code>         | Command joint-space target position.                             | manual motion |
| <code>movePosPath(path, vPath, tPath)</code>     | Command sequence of joint-space waypoints.                       | manual motion |
| <code>moveVel(arm_vel, a=10, t=0)</code>         | Command joint-space velocity.                                    | manual motion |
| <code>moveEEPoint(pose, v=50, t=0)</code>        | Move end-effector to cartesian pose (point mode).                | manual motion |
| <code>moveEEPointPath(path, vPath, tPath)</code> | Move through end-effector pose waypoints (point mode).           | manual motion |

| API                                                           | Description                                                            | Group         |
|---------------------------------------------------------------|------------------------------------------------------------------------|---------------|
| <code>moveEELine(pose, v=50, t=0)</code>                      | Move end-effector to cartesian pose (line mode).                       | manual motion |
| <code>moveEELinePath(path, vPath, tPath)</code>               | Move through end-effector pose waypoints (line mode).                  | manual motion |
| <code>moveToolPoint(pose, v=50, t=0)</code>                   | Move tool frame to cartesian pose (point mode).                        | manual motion |
| <code>moveToolPointPath(path, vPath, tPath)</code>            | Move tool frame through cartesian waypoints (point mode).              | manual motion |
| <code>moveToolLine(pose, v=50, t=0)</code>                    | Move tool frame to cartesian pose (line mode).                         | manual motion |
| <code>moveToolLinePath(path, vPath, tPath)</code>             | Move tool frame through cartesian waypoints (line mode).               | manual motion |
| <code>goHome(v=50, t=0)</code>                                | Move robot arm to predefined home pose.                                | manual motion |
| <code>moveGripperPos(pos, v=50, t=0)</code>                   | Command gripper joint position.                                        | manual motion |
| <code>startTeach(), stopTeach()</code>                        | Enter or exit teach mode.                                              | teach mode    |
| <code>startPlayback(), stopPlayback(), resetPlayback()</code> | Control playback mode lifecycle.                                       | playback mode |
| <code>getPos(), getVel(), getTor()</code>                     | Read current joint position, velocity, and torque.                     | feedback      |
| <code>getEEPose(), getToolPose()</code>                       | Read current end-effector/tool cartesian pose.                         | feedback      |
| <code>isArmJointEnabled()</code>                              | Read per-joint arm enable state.                                       | status        |
| <code>getStatus(), printStatus(status)</code>                 | Read or print aggregated runtime status ( <code>SystemStatus</code> ). | status        |
| <code>getConfigurator(), getKinematics()</code>               | Access specialized subsystems for config/kinematics.                   | subsystems    |

## Configuration API (`Configurator`)

| API                                                 | Description                                           |
|-----------------------------------------------------|-------------------------------------------------------|
| <code>setToolOffset(offset), getToolOffset()</code> | Set or get tool frame offset from end-effector frame. |

| API                                                                                                  | Description                                     |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <code>setArmPosLimits(limit), setArmVelLimits(limit)</code>                                          | Set joint position/velocity safety limits.      |
| <code>getArmPosLimits(), getArmVelLimits()</code>                                                    | Get active joint safety limits.                 |
| <code>getArmMaxPos(), getArmMaxVel()</code>                                                          | Get hardware max position/velocity constraints. |
| <code>getArmMaxPosJump(), getArmMaxVelJump()</code>                                                  | Get anti-jump protection limits.                |
| <code>getArmMaxPosFollowError(),<br/>getArmMaxVelFollowError(),<br/>getArmMaxTorFollowError()</code> | Get following-error thresholds.                 |
| <code>setGripperPosLimits(limit), getGripperPosLimits()</code>                                       | Set or get gripper position limits.             |
| <code>getGripperMaxPos()</code>                                                                      | Get gripper hardware max range.                 |

## Kinematics API (**Kinematics**)

| API                                                   | Description                                                                |
|-------------------------------------------------------|----------------------------------------------------------------------------|
| <code>forwardKinematics(arm_pos)</code>               | Compute cartesian pose from 6-joint arm state.                             |
| <code>inverseKinematics(ee_pose,<br/>arm_pos)</code>  | Solve 6-joint arm state for a target cartesian pose; returns <b>bool</b> . |
| <code>eulerToQuaternion(euler)</code>                 | Convert intrinsic XYZ Euler angles to quaternion.                          |
| <code>quaternionToEuler(quat)</code>                  | Convert quaternion to intrinsic XYZ Euler angles.                          |
| <code>eeToToolPose(ee_pose,<br/>tool_offset)</code>   | Transform end-effector pose into tool pose.                                |
| <code>toolToEEPose(tool_pose,<br/>tool_offset)</code> | Transform tool pose into end-effector pose.                                |

## Types (**types.h**)

| Type / Enum                                             | Description                                                       |
|---------------------------------------------------------|-------------------------------------------------------------------|
| <b>Position</b> , <b>Quaternion</b> , <b>EulerAngle</b> | Cartesian position, orientation, and intrinsic XYZ Euler angles.  |
| <b>Pose</b>                                             | Cartesian pose = <b>Position</b> + <b>Quaternion</b> orientation. |
| <b>JointState6f</b> / <b>JointState1f</b>               | 6-DOF arm joint values / 1-DOF gripper value.                     |
| <b>JointState6b</b> / <b>JointState1b</b>               | 6-DOF arm enable flags / gripper enable flag.                     |

| Type / Enum                                   | Description                                                                                                                                                          |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JointState6u / JointState1u                   | Per-joint diagnostic bitmasks for arm and gripper.                                                                                                                   |
| Path<T>                                       | Waypoint sequence ( <code>std::vector&lt;T&gt;</code> ).                                                                                                             |
| RobotJointStatef / RobotJointStateb           | Combined arm+gripper state in float or bool forms.                                                                                                                   |
| JointLimits<T>, JointLimits6f / JointLimits1f | Per-joint min/max limits for arm and gripper.                                                                                                                        |
| SmoothingMethod                               | <code>kLinear</code> , <code>kCos</code> , <code>kCubic</code> , <code>kQuintic</code> , <code>kNone</code> .                                                        |
| RobotState                                    | <code>kStartup</code> , <code>kIdle</code> , <code>kMoving</code> , <code>kSettling</code> , <code>kError</code> , <code>kRecovery</code> , <code>kShutdown</code> . |
| PlanResult                                    | <code>kSuccess</code> , <code>kPoseNotReachable</code> , <code>kLinearPathFailed</code> .                                                                            |
| SystemStatus                                  | Aggregated status: <code>robot_state</code> , <code>plan_result</code> , <code>robot_diagnostic_flags</code> , per-joint diagnostic flags.                           |
| DiagnosticFlags                               | Bitmask flags for target/actuator saturation, jump limits, boundary safety, planner events, and hardware/tracking faults.                                            |